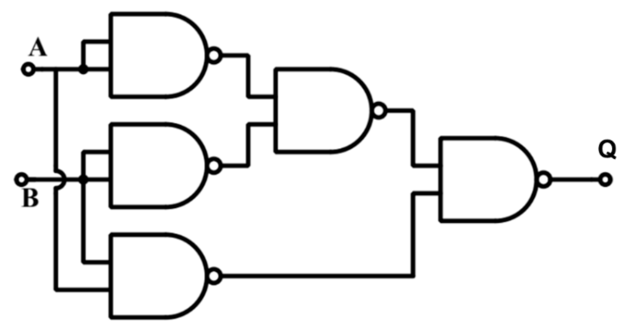
Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Semester II, 2025-2026

Tutorial-11 (Computation and Storage elements in Processors)

Q1. We have both 2-input NAND gates and 2-input NOR gates available.



A. You are given the following logic circuit that has been implemented with 2-input NAND gates. Write its truth table and find its functionality.

B. If you were to implement the functionality given in Q.1A using 2-input NOR gates, draw the logic circuit using the least number of gates.

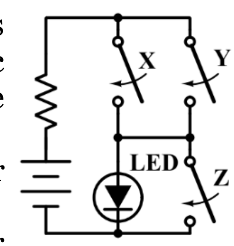
Q2. A literal is a single variable in a term, in complemented or un-complemented form. Simplify the following Boolean expression to minimum number of literals.

a) $(a + b + \overline{c}) \cdot (\overline{a} \cdot \overline{b} + c)$

b) $\overline{a} \cdot b \cdot c + a \cdot b \cdot \overline{c} + a \cdot b \cdot c + \overline{a} \cdot b \cdot \overline{c}$

c) $\overline{(x + y)} \cdot (\overline{x} + \overline{y})$

Q3. Each switch in the circuit diagram on the left is closed whenever its controlling variable (X, Y, Z) is TRUE (Logic 1) and open when FALSE (Logic 0). Let the output function L (X, Y, Z) = 1 when the LED is lit, and 0 when the LED is dark.

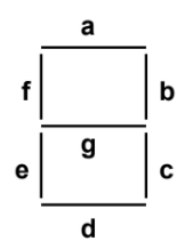


A. What is the logic function of X, Y and Z which will be TRUE whenever the LED is lit?

B. What is the logic function of X, Y and Z which will be FALSE whenever the LED is dark?

Q4. We want to display a decimal digit using a seven segment display. Decimal digits are expressed using 4 bits A, B, C and D; with A as the most significant bit. Combinations of ABCD representing 10, 11, 12, 13, 14 and 15 are not used and can be considered as 'don't care'.

A seven segment display has segments 'a' to 'g' as shown in the figure on the right. Combinations of these segments light up to display different decimal digits.



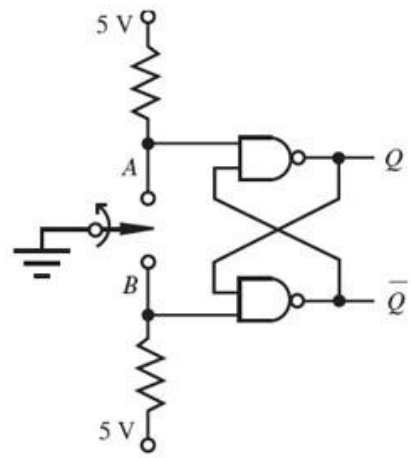
A. The 'a' segment should light up for digits 0, 2, 3, 5, 6, 7, 8 and 9. Make a K-map for a function of ABCD which will be TRUE when the 'a' segment should be lit. Mark "don't care" entries as X.

B. Show the maximal groupings of ‘1’s (including X’s when convenient to take it as ‘1’) in the K-map above and derive the minimal logic expression in terms of ABCD for lighting up segment ‘a’.

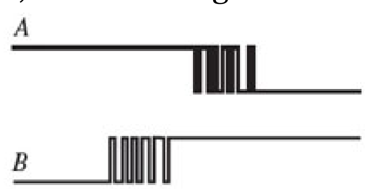
Q5. You are learning to design a basic processor where the CPU needs to communicate with four separate memory registers (Registers 0, 1, 2, and 3). To select which register it wants to talk to, the CPU outputs a 2-bit address (A1, A0).

A. To write data, the CPU needs to send a single 1-bit Data_Write signal to one specific register based on the 2-bit address. Design and draw the logic circuit for this using only AND gates and NOT gates.

B. Now, the CPU needs to *read* data. The four registers are sending out four continuous data signals (D0, D1, D2, D3), but the CPU only has one Data_Read input pin. Design and draw the logic circuit that uses the CPU's address lines (A1, A0) to read the correct register's data into the CPU's single Data_Read pin. Write the Boolean expression for Data_Read and deduce to it to build the circuit with only 2-input NAND gates.



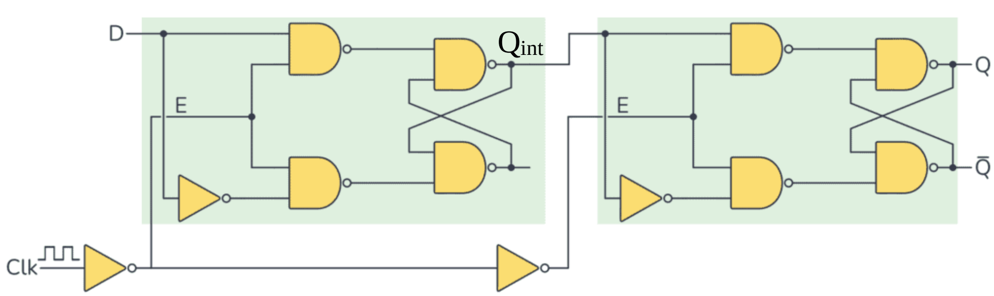
Q6. Mechanical switches rarely provide a clean logic transition. When you press a button (e.g. on a music player), the metal contacts physically "bounce," sending a rapid, noisy burst of 1s and 0s into the circuit before finally settling. In this question, when the switch is moved from B to A, the following A and B waveforms are generated.

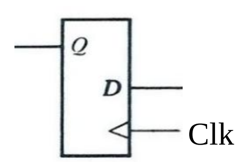


A. Write the characteristic truth table for the given circuit.

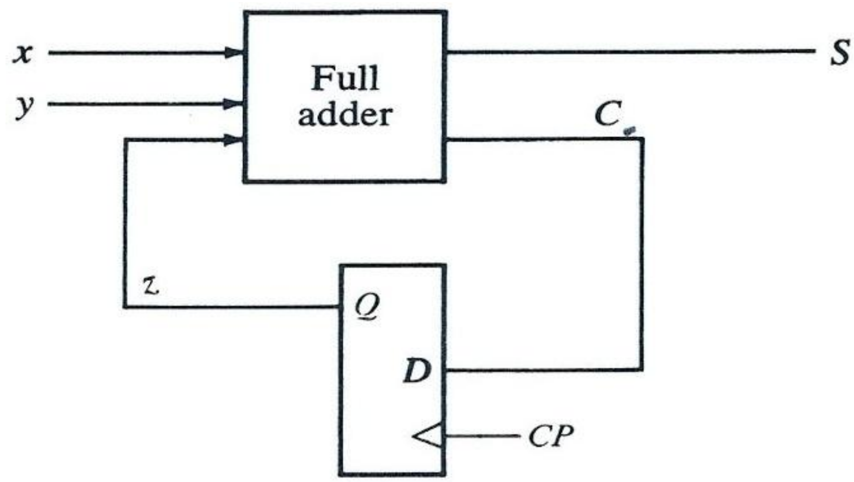
B. Draw the corresponding the output Q. Explain how this circuit solves the problem of switch de-bouncing.

Q7. In the logic circuit given below (with its symbol), if the input D changes to 1 while Clk = 0, what is the immediate logic level at the intermediate node Q_{int}, and what specific clock transition must occur for that value to finally appear at the output Q?





Q8. A sequential circuit has one flip-flop, Q; two inputs, x and y; and one output, S. It consists of a full-adder circuit connected to a D flip-flop, as shown in the figure. Derive the state table and state diagram of the sequential circuit.



More Practice Problems

- Digital Systems Book by Tocci
- Digital Design Book by Maurice Mano